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Short Communication

A comprehensive study into false positive rates for 'other' biological samples using common presumptive testing methods

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ABSTRACT

The identification of biological fluids or materials in forensic samples is a key requirement in forensic science that relies on chemical and biological based tests, most of which exhibit false positivity. When reporting results from such tests, Forensic Scientists use words such as probable, possible, and likely, without always being able to provide robust support for these conclusions. In collating information about false positive rates for a number of these tests, we found limited research into the cross reactions observed from 'other' biological samples in commonly encountered case sample stains. By 'other' we mean biological fluids or materials that are not the primary target of the presumptive test being used. Here we carry out a specificity study to fill gaps in the literature for a number of the presumptive chemical, biological and immunochromatographic tests used to presumptively screen for blood, semen and saliva. The tests selected for this study are the widely used tests: Luminol, TMB/Combur^3 Test® E, Kastle-Meyer (KM), RSID™ - Blood, ABAcard® HemaTrace®, Acid Phosphatase (AP), ABAcard® p30, RSID™ - Semen, Phadebas® 'Tube' Test, Phadebas® 'Press' Test, and RSID™ - Saliva tests. Specificity for each of these was tested in known samples, from volunteers, of blood, semen, saliva, urine, sweat, vaginal material, faeces and breast milk, and then false positive rates were determined.

1. Introduction

Biological fluid identification (BFID) is a crucial part of forensic investigation, assisting investigators in the reconstruction of alleged or unknown events. The results of body fluid testing can be incorporated into Bayesian Networks (BNs), providing an evaluation of source in a robust framework [1–6]. Commonly used in most laboratories are tests which are often presumptive, meaning that false positive, or negative results can occur [1,7–9]. False positive reactions can be caused by cross reactivity with other body fluids and with other substances such as food or plant material [10–12]. Whilst studies on "other substances" are widely available, some of which report the impact of different biological sources on false positive rates [13–25]. Many studies are not relevant, are unable to be applied to the tests being used or there is insufficient data available to support the estimation of false positive rates, for inclusion in reporting frameworks.

We found that as most blood presumptive tests react to the presence of hemoglobin, which is only present in blood, there is a lack of documented research performed on other biological samples which may or may not react similarly. For example, we were unable to find any literature regarding the testing of TMB/Combur^3 Test® E on any biological samples other than blood. We were unable to find literature with regards to any biological samples other than blood, semen or saliva for both luminol and Kastle-Meyer (KM). In contrast to this, as it is known that acid phosphatase (AP), prostate specific antigen (p30) and α-amylase are present in lower concentrations in multiple biological fluids or materials, many of the tests for semen and saliva have been more thoroughly evaluated, with the exception of the RSID™ - Semen test on sweat and the Phadebas® 'Press' test on breast milk.

In this paper, we focus on a selection of tests for locating and identifying blood, semen, and saliva and test them on known samples of blood, semen, saliva, urine, sweat, vaginal material, faeces and breast milk. Blood, semen, saliva, and vaginal material were selected for testing as they are often associated with sexual assault cases. Urine, sweat and faeces were selected as they may be present in commonly sampled areas like underwear. Breast milk was chosen as it has been known to give

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false positives for some semen and saliva presumptive tests [22,26]. The tests used in this study were: luminol [27,28], TMB/Combur^3 Test® E [29], Kastle-Meyer (KM) [30], RSID™ – Blood [13], ABAcard® HemaTrace® [23], Acid Phosphatase (AP) [31], ABAcard® p30 [15], RSID™ - Semen [18], the Phadebas® 'Tube' Test [20], the Phadebas® 'Press' Test [24], and RSID™ – Saliva [22]. These tests were selected as, with exception of the RSID™ kits, they are all of the current tests used in forensic casework in our laboratory. The RSID™ kits were included as it has been suggested that they may be suitable alternatives to some of the current testing methods being used.

2. Materials and methods

2.1. Collection of biological samples

Biological samples were collected with informed consent from volunteers with ethics approval from the Auckland Health Research Ethics Committee (AHREC; reference number AH22404). Each participant provided four samples each of sweat, faeces and vaginal material, and one sample each of blood, semen, saliva, urine and breast milk where relevant and possible. Some samples of breast milk were frozen after collection prior to being used in this study. Samples were stored at 2–8 °C prior to testing.

Aliquots of 50 μL from each of the semen, saliva, urine, and breast milk samples were deposited on four separate 3 cm^2 white cotton fabric swatches. For blood, only one sample was prepared with a single 50 μL aliquot which was then divided into four equal parts to ensure a similar ratio of fluid to substrate as the other biological samples. Samples were left to air dry at room temperature for 1 h before testing.

Four sweat samples were collected from each participant on 3 cm^2 white cotton fabric swatches after exercise, two from the forehead and two from the underarms. Faecal samples were collected via used toilet paper, sampled to contain approximately the same amount of sample. Female participants provided four vaginal swabs.

A total of 21 participants (18 female and 3 male) donated the following biological samples: 18 samples of blood, 3 samples of semen, 20 samples of saliva, 14 samples of sweat, 5 samples of faeces, 17 samples of urine, 9 vaginal swabs and 2 samples of breast milk, resulting in a total of: 72 samples of blood, 12 samples of semen, 80 samples of saliva, 56 samples of sweat, 20 samples of faeces, 68 samples of urine, 36 vaginal swabs and 8 samples of breast milk for testing.

Positive and negative controls were performed for all tests. The target fluids (blood, semen or saliva) were used as positive controls for their corresponding tests. The methods and substrates used for the positive and negative controls mirrored those of the samples for each body fluid and test.

To minimize variability between tests, the extraction methods for both ABAcard® kits (HemaTrace® and p30), as well as the RSID™ kits (Blood, Semen and Saliva) were modified to be performed together on a common extract. Within the volume limitations of the manufacturer's protocols, this allowed the samples to be comparable as aliquots of the same extract were tested by all five kits. Initial experiments (data not shown) found that the RSID™ - Universal Buffer present in the RSID™ kits could be used for all kits tested. The volume of RSID™ - Universal Buffer required to perform all five tests, including initial extraction of the samples, dilution before application and the volume applied to the test strip was determined. It was found that due to the volume of liquid required to accurately perform the Phadebas 'tube' test, it was to be excluded from the combined extractions.

It is acknowledged that any modifications or variations to the methodology may affect the results obtained, and that if different methods are carried out different results may be obtained. This should be considered by any laboratory wishing to use the results from this study.

We explored the use of Tris-EDTA (TE) buffer to prepare samples, however all RSID™ kits gave false positive reactions for negative controls and this was therefore discontinued.

2.2. Tests for blood

The luminol test was performed according to the method described in [28]. Combur^3 Test® E strips (Roche Diagnostics International AG Rotkreuz, Switzerland) were applied directly to each stain and then moistened with sterile water. Positive results were recorded by the development of a dark blue/green colour after 1 min.

A two-step Kastle-Meyer test was used according to Cox 1991 [32]. A positive result was recorded following the development of a deep pink colour within the first few seconds of the application of the second step. Any colour developed prior to this was recorded as having an oxidizing agent present.

To test biological samples with the RSID™ - Blood (Independent Forensics, Lombard IL, USA) method, a 3 mm^2 cutting or ~ 1/3 of the vaginal swab head was extracted in 300 μL of RSID™ - Universal Buffer for 1 h at room temperature. 50 μL of extract was added to a further 50 μL of RSID™ - Universal Buffer before application of 100 μL to the sample well of the RSID™ - Blood cassette. For testing with the ABAcard® HemaTrace® method, (Abacus Diagnostics, Inc., West Hills, CA, USA) a 200 μL of a stain extract (total extract 300 μL from a 3 mm^2 cutting) was applied directly to the sample window labelled 'S' of a ABAcard® HemaTrace® test device. For each test, results were recorded after 10 min.

2.3. Tests for semen

The AP test was used as described by Virkler and Lednev in 2009 [7]. A dampened filter paper was applied to each sample under moderate pressure then brushed with reagent. A positive result was recorded after the development of a strong purple colour within 2 min of application of the reagent.

For testing with the ABAcard® p30 method (Abacus Diagnostics, Inc., West Hills, CA, USA), a 3 mm^2 cutting or ~ 1/3 of the vaginal swab head was extracted in 300 μL of RSID™ - Universal Buffer for 1 h at room temperature and then 200 μL was applied directly to the sample window.

For testing using the RSID™ - Semen (Independent Forensics, Lombard IL, USA) method, a 3 mm^2 cutting or ~ 1/3 of the vaginal swab head was extracted in 300 μL of RSID™ - Universal Buffer for 1 h, at room temperature. 50 μL of extract was added to a further 50 μL of RSID™ - Universal Buffer before application of 100 μL to the sample well. For both the ABAcard® p30 and RSID™ - Semen, results were recorded after 10 min.

2.4. Tests for saliva

The Phadebas® 'tube' test was used as described by Phadebas AB (Phadebas AB, Kristianstad, Sweden) [33] with some modifications. A 3 mm^2 cutting of each sample was extracted in 1 mL RSID™ - Universal Buffer. 450 μL was taken and mixed with ~ 10 mg of a crushed Phadebas tablet and incubated at 37 ± 2 °C for 15 min. The time of appearance and strength of the blue colour in the supernatant was measured. A darker blue colour indicates a stronger positive reaction. Samples were compared to a prepared standard curve (fresh saliva diluted to 1:10, 1:100 and 1:1,000) to quantify the strength of the reaction.

The Phadebas® 'press' test was performed according to the method described by Willott and Griffiths [34]. Phadebas® paper (Phadebas AB, Kristianstad, Sweden) was applied directly to each stain then moistened with sterile water. Pressure was applied and positive results were recorded with the development of a blue colour on the test sheet at regular intervals of 10 min up to a maximum of 40 min.

A 3 mm^2 cutting or ~ 1/3 of the vaginal swab head was extracted in 300 μL of RSID™ - Universal Buffer for 1 h, at room temperature. 50 μL of extract was added to a further 50 μL of RSID™ - Universal Buffer

before application of 100 μL to the sample well window of the RSID™ - Saliva (Independent Forensics, Lombard IL, USA) cassette. Results were recorded after 10 min.

3. Biological fluid identification results

False positive rates were calculated using the total number of positive reactions over the total number of observations and presented as a percentage.

In most cases samples were divided into four to create four new sample or four samples were taken from the same person at the same moment. This data shown is the average over all of the tests carried out, including inter and intra variability. Where the samples were divided into four, we were able to assess repeatability, however these replicate samples are not independent of each other. This has been considered in the results provided below for each test and should be taken into account when considering using this data.

For the purposes of this study, we have defined a weak positive as a slower developing reaction with a less intense colour change.

3.1. False positive rates of the tests for blood

The results obtained from the testing of each sample using the presumptive tests for blood are presented in Fig. 1a–d. Note: a figure has not been generated for RSID™ - Blood. A full summary of all of the results obtained, including where no false positive reactions were observed, can be found in the supplementary data in Appendix A. The luminol test gave the highest number of false positives compared to the other four blood presumptive tests evaluated in this study. At least one of every biological sample tested gave a false positive result (weak or strong) for blood, including 79 % of saliva samples and 50 % of semen samples. Saliva and semen also gave high positive reaction rates with one other blood presumptive test, ABAcard® HemaTrace®.

When using the Combur^{3} Test® E strips, the only strong positive reactions observed were for three of the saliva samples. In addition, semen (1 sample), saliva (3 samples), vaginal material (8 samples) and 90 % of faeces (18 samples) gave a weak positive result. Faeces was the only biological source, when tested for the presence of blood, to give a positive reaction with the Combur^{3} Test® E strips and negative for all other blood presumptive tests. This occurred with two separate participants for all faeces samples provided.

No samples tested with KM gave a strong positive result. Only three biological sources resulted in a weak positive reaction: saliva, faeces, and vaginal material.

No false positive results were obtained from any of the testing of samples using RSID™ - Blood. This test is the only blood presumptive test designed to detect glycophorin A as an alternative to haemoglobin.

The testing of semen samples with ABAcard® HemaTrace® resulted

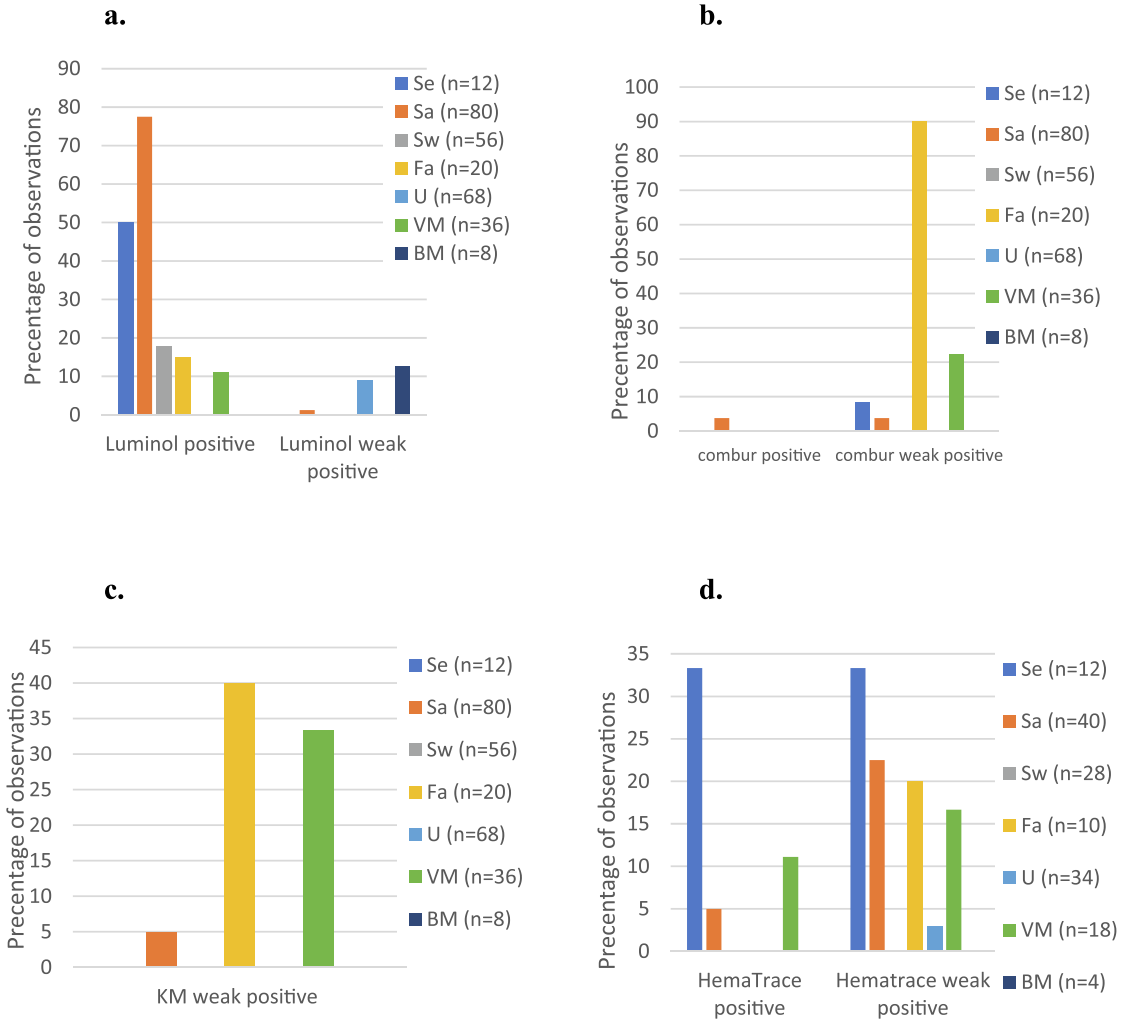


Fig. 1. False positive rates of presumptive tests for blood from human biological sources. Results are shown for biological samples from semen (Se), saliva (Sa), sweat (Sw), faeces (Fa), urine (U), vaginal material (VM) and breast milk (BM) using, in (a.), the luminol test, in (b.), Combur3 Test® E (Combur), in (c.), the Kastle Meyer (KM) test, and in (d.), the ABAcard® HemaTrace® (HemaTrace) test. Negative test results have not been presented.

in 66 % of samples giving a positive reaction (33 % strong positive and 33 % weak positive). Saliva (11 samples), faeces (2 samples), urine (1 sample) and vaginal material (5 samples) also gave positive reactions with ABAcard® HemaTrace®.

Other than for luminol, all of the sweat and breast milk samples gave a negative result for each of the blood presumptive tests.

3.2. False positive rates of the tests for semen.

The results obtained from the testing of each sample using the presumptive tests for semen are presented in Fig. 2a. & b. Note: a figure has not been generated for RSID™ - Semen. A full summary of all of the results obtained, including where no false positive reactions were observed, can be found in the supplementary data in Appendix A. All samples of faeces gave a positive reaction in the AP test, 80 % giving a strong positive and 20 % giving a weak positive. All male urine (8 samples) tested gave a weak positive result for AP. This result was not unexpected due to the nature of male anatomy and physiology. Two female participants (6 urine samples) gave weak positive result for AP.

55 % of vaginal material samples gave a positive result (4 strong positive results all from one female participant and 16 weak positive results, 4 each from four female participants) for AP. The four strong positive results were all obtained from one female participant who had stated they had sexual intercourse 4 days prior to sample collection, which may account for these results. Of the 16 weak positive results, these were observed from 4 of the female participants. Three of these females (accounting for 12 samples) had no documented sexual contact within the 10 days prior to sample collection. The fourth female had stated sexual intercourse 8 days prior to sample collection.

The urine samples collected from one male donor (2 samples) gave weak positive results for p30. This result was not unexpected due to the nature of male anatomy and physiology. In addition, one urine sample from a female participant also resulted in a weak positive for p30. The urine sample from this female participant did not give an AP positive result.

No false positive results were obtained from any of the testing of samples using RSID™ - Semen. This test is used to identify the protein semenogelin in semen which is responsible for the clotting during ejaculation [35]. Note: In Fig. 2a. & b. (below) a figure has not been generated for RSID™ - Semen.

3.3. Results for saliva presumptive tests

The results obtained from the testing of each sample using the presumptive tests for saliva are presented in Fig. 3a., b. & c. A full summary of all of the results obtained, including where no false positive reactions were observed, can be found in the supplementary data in Appendix A. All but one faeces sample gave a positive reaction for the Phadebas® ‘tube’ test. 80 % of faeces samples gave a strong positive reaction and 15 % of samples gave a weak positive reaction. All other biological sources tested gave a negative result.

The Phadebas® ‘press’ test gave the highest number of false positives compared to any of the other presumptive tests in this study, including the other two saliva presumptive tests evaluated. All faeces and breast milk samples resulted in a positive reaction with majority of samples giving a strong positive (90 % faeces and 88 % breast milk). Vaginal material also gave a high positive reaction rate for the Phadebas® ‘press’ test. Out of the biological samples that produced positive results in the Phadebas® ‘press’ test, sweat gave the lowest positive reaction rate of 17 %.

Like the Phadebas® ‘press’ test, all faeces and breast milk samples gave a positive reaction for RSID™ - Saliva. 50 % of vaginal material and 33 % of semen samples gave a positive reaction (all weak positives). Additionally, 14 % of sweat (4 samples) and 3 % of urine (1 sample) tested gave a positive reaction. All blood samples gave negative results in the tests for saliva.

In addition, six samples comprising two mixtures of 50 μL each of blood and semen, blood and saliva, and semen and saliva, were prepared and tested using all presumptive tests. No interference was observed with the any of the mixture samples. Each mixture tested positive for the expected components and negative for the third.

4. Discussion

4.1. Presumptive tests for blood

The luminol test was found to be the least specific; reacting positively to at least one of every biological sample tested. Positive results obtained from saliva samples may have occurred due to gingivitis, mouth sores or other common ailments resulting in bleeding gums. Some of the highest rates of positive reactions occurred with semen and saliva samples and corresponded with results obtained from Combur^{3} Test® E strips, KM, and ABAcard® HemaTrace® tests. These results are in contradiction to Tobe et al.’s previous study [25] in which all semen samples reacted negatively with luminol and all positively with KM. One explanation for this may be the longer time between administering the KM test and recording the results used in the earlier study. An explanation for the positive reactions recorded for semen with luminol and also with

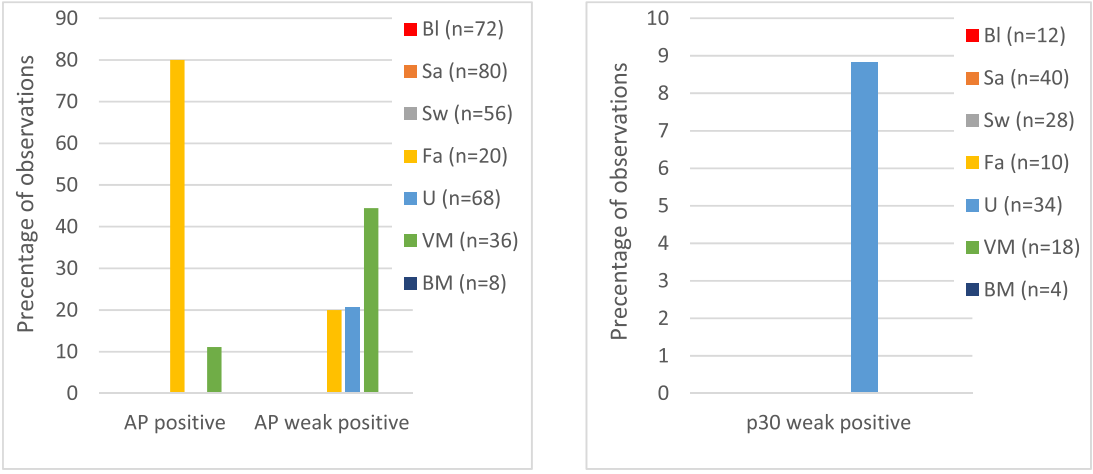


Fig. 2. false positive rates of presumptive tests for semen from human biological sources. Results are shown for biological samples from blood (Bl), saliva (Sa), sweat (Sw), faeces (Fa), urine (U), vaginal material (VM) and breast milk (BM) using, in (a.), the acid phosphatase (AP) test and, in (b.), the ABAcard® p30 (p30) test. Negative test results have not been presented.

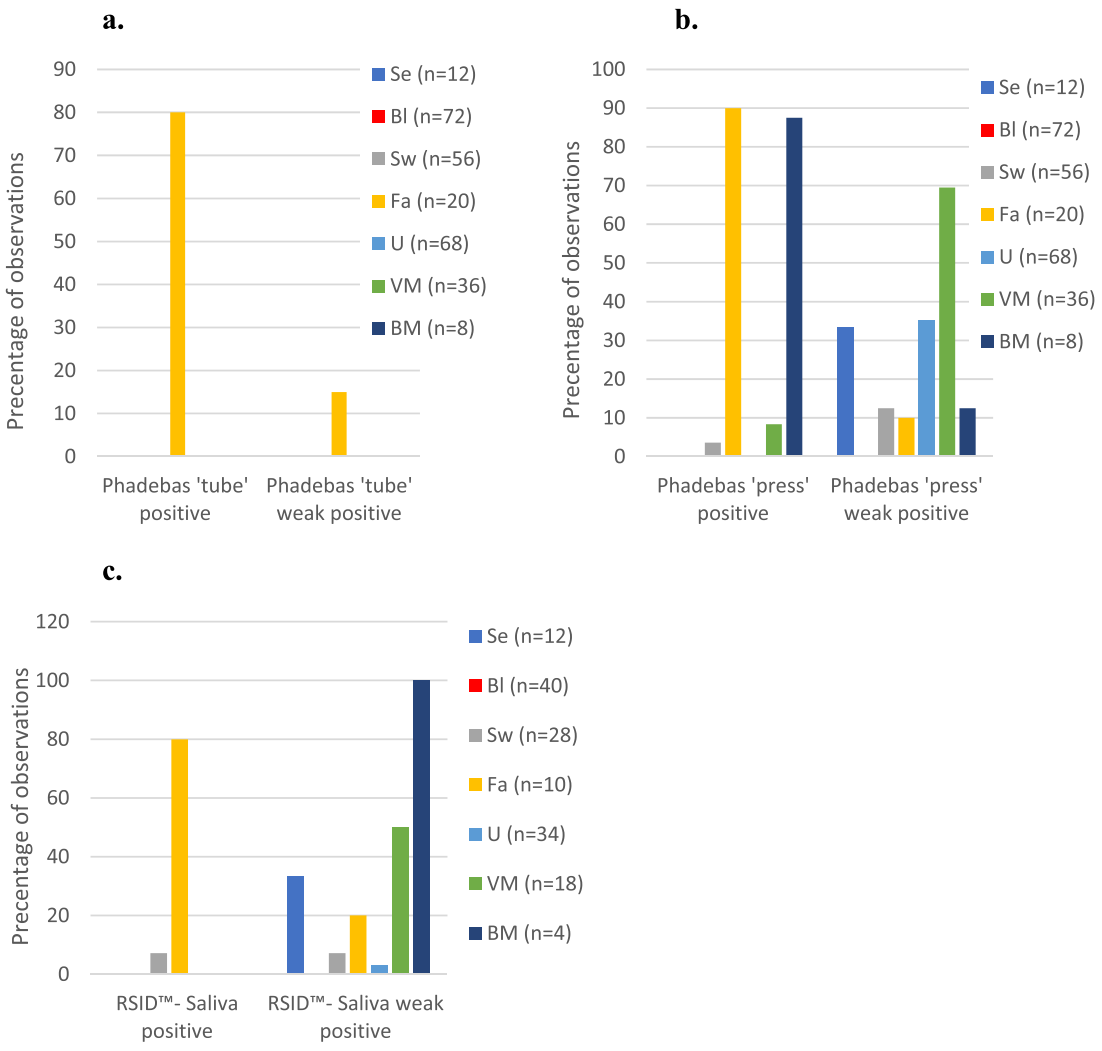


Fig. 3. False positive rates of presumptive tests for saliva from human biological sources. Results are shown for biological samples from blood (Bl), semen (Se), sweat (Sw), faeces (Fa), urine (U), vaginal material (VM) and breast milk (BM) using, in T**able 3a., the Phadebas® 'tube' test and the Phadebas® 'press' test and, in T**able 3b., the RSID™ - Saliva test. Negative test results have not been presented.

ABAcard® HemaTrace® is that trace levels of both copper and iron [36,37] are both known to give false positive results in tests based on the presence of haemoglobin [32]. Luminol was the only blood presumptive test to give a positive reaction to sweat (10 samples) and breast milk (1 sample) in this study. Note that specificity studies for sweat and breast milk have not been reported previously.

Faeces and vaginal material also gave positive results in multiple blood presumptive tests: luminol, Combur^{3} Test® E strips, KM, and ABAcard® HemaTrace®. These results are supported by Johnston et al. [23] for ABAcard® HemaTrace® testing. No other specificity studies regarding the blood presumptive tests discussed here included data for faeces or vaginal material. These results are not unexpected as both faeces and vaginal material can contain trace levels of blood. These results show that faecal staining, which is likely to appear red/brown in colour could be mistaken for blood.

As expected, the Combur^{3} Test® E test was shown to be less specific than the two-step KM test, which was shown to have similar specificity to ABAcard® HemaTrace®.

Luminol (6 samples) and ABAcard® HemaTrace® (1 sample) both reacted positively to at least 1 urine sample. This is in contradiction to a previous study [23] which reported all negative results for blood when testing urine with ABAcard® HemaTrace®. This may be explained by the larger number of samples tested in our study. We tested 34 samples from 17 participants whereas the previous study only tested samples from four participants.

RSID™ - Blood was determined to have the highest specificity of any test as it did not react positively to any biological sample other than blood. This may be attributed to the test's identification of glycophorin A, as opposed to hemoglobin. Our results obtained from semen, saliva, urine and breast milk samples are in agreement with Schweers at al 2007 [13]. On the contrary, Horjan et al. 2016 [14] reported 10 out of 20 saliva samples reacting positively. This could be explained by the presence of blood in the samples tested in either this study or the earlier study.

4.2. *Presumptive tests for semen*

The AP test was the least specific test for semen as acid phosphatase is commonly found in nature and in other biological samples [1,7,8]. The AP test yielded positive results for faeces (all 20 samples), urine (8 out of 12 samples from males and 6 out of 56 samples from females) and vaginal material (20 samples). Some of the positive results observed in the vaginal material samples collected from females can be accounted for by some of the sexual history data collected at the time of sampling (see Section 3.2). The AP results we obtained support findings in previous specificity studies [38–40]. Breast milk did not react positively for AP in this study or other previously completed studies [41,42]. In previous studies, blood, saliva, and sweat also reacted positively to AP

[39,41,42]. Different testing procedures may account for contrary results obtained for the blood samples. Hooft et al. yielded 11 positive blood samples (plus another 11 serum samples and 2 plasma samples) from two studies, and a total of 170 tests, one performed in 1990 and the other in 1994. In these two studies, AP was tested using a strip infused with AP reagent pressed onto a moistened stain. In our study, and in the more recent specificity studies performed for the detection of AP, a piece of filter paper was pressed into the moistened sample and then the AP reagent was directly applied to the filter paper [41,42].

ABAcard® p30 presumptive testing was shown to give high levels of specificity for semen. Urine was the only biological sample that resulted in a positive reaction (three samples) in this study. Two positive urine samples were provided by one of the male participants; therefore, these results were not unexpected. The single female urine sample appears to be an outlier in this data set and is unexplained. Whilst we can't explain why this occurred, the remaining three samples from this female were negative for p30, and all four samples from this female were negative for AP, and this is why we consider it an outlier.

RSID™ - Semen was determined to have the highest specificity as it did not react positively to any biological sample other than semen. This may be attributed to the test's identification of semenogelin, which is only present during the clotting stage of male ejaculation. In addition, semenogelin has yet to be found present in any pre-ejaculatory fluids which may account for its high specificity for semen in regards to male urine [43,44]. The results found in this study are in complete agreement with previous specificity studies performed on RSID™ - Semen [16,18,45]. Note that semenogelin has previously been proposed as a confirmatory mRNA marker for forensic application [46]. Additionally, these test results align with the manufacturer's claim which guarantees no cross-reactivity with human blood, saliva, urine and breast milk [18].

4.3. Presumptive tests for saliva

The Phadebas® 'press' test gave the lowest specificity results for the saliva presumptive tests, yielding positive results for every biological sample tested except blood including all faecal and breast milk samples tested. Although breast milk has not been tested in any previous specificity studies for saliva, all other results obtained for this test are supported by data obtained from literature [24,46 47]. Positive results for α-amylase in semen were not unexpected, as the enzyme can be found in semen at low concentrations [21].

Of all the RSID™ testing kits, RSID™ - Saliva has the lowest specificity despite the manufacturers claim of no cross-reaction with human blood, semen, urine, and vaginal material [22]. In this study, RSID™ - Saliva reacted positively to semen, sweat, faeces, urine, vaginal material, and breast milk which is in direct contradiction to the manufacturer's claim. Most of the results are supported by data collected in previous specificity studies which has shown false positive reactions from semen, sweat, faeces, urine and breast milk, but not vaginal material [48,49]. Based on our findings, RSID™ - Saliva should not be considered confirmatory for the presence of saliva.

The Phadebas® 'tube' test appeared to be the most specific for saliva. Only faecal samples reacted positively, contrary to data obtained from previous studies where positive reactions were obtained from faeces as well as urine, vaginal material, breast milk and semen [20,21].

5. Concluding remarks

Whilst we know false positive reactions can also be caused by cross reactivity with other substances such as food or plant material, studies on these other substances are widely available. Therefore, the purpose of our study was to "fill the gaps" in existing literature relating to the use of presumptive tests exhibiting false positivity, by expanding the range of biological samples tested compared to previous studies in a systematic way. Specificity for each of these was tested in known samples of blood, semen, saliva, urine, sweat, vaginal material, faeces and breast milk and false positive rates were determined.

Although we did not detect any false positives in our study on RSID™-Blood we are aware of an earlier study where saliva samples tested positive [14] with this test, although this may be due to the presence of blood in the saliva samples. There are no reported examples of false positive rates for RSID™-Semen in the literature. We identified, that when performed as described here, our findings could be viewed as supportive of previous recommendations for the use of a semenogelin based test for the identification of semen [45,46]. We determined that most other tests did have a false positivity rate.

Some of the results presented here, for tests other than RSID™ - Semen, are contrary to previous studies, which may be due in part to earlier and or different data collection practices used. This highlights one challenge of the need to incorporate data from the existing literature when assessing forensic evidence. These false positivity rates can be used to inform the reporting of forensic evidence and could be included in reporting frameworks, such as Bayesian Networks, to improve the evaluation of forensic evidence.

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CRediT authorship contribution statement

Angela Stroud: Methodology, Validation, Investigation, Writing – original draft, Writing – review & editing. **Amelia Gamblin:** Conceptualization, Methodology, Writing – review & editing, Supervision. **Penny Birchall:** Investigation, Writing – review & editing. **SallyAnn Harbison:** Conceptualization, Methodology, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Stephanie Opperman:** Conceptualization, Methodology, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scijus.2023.04.006.

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